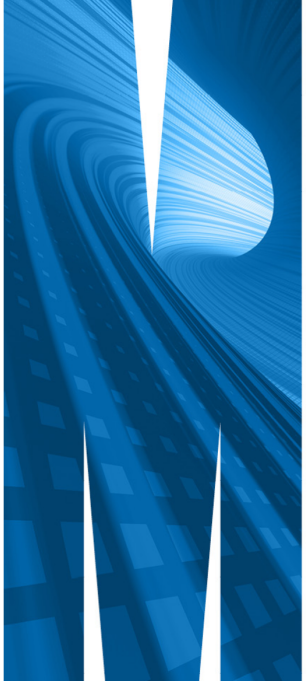
MONASH University

ETX1100/ETX5900
Business Statistics

Week 2: Probability and distributions

1 December 2026



Outline

1 Probability language

2 Contingency tables

3 Continuous probability distributions

4 Integrated application

Today's journey

1. Translate business uncertainty into events and probabilities.
2. Build contingency tables and distinguish joint, marginal, and conditional probability.
3. Assess whether two events are independent.
4. Use Normal and Student's *t* distributions in Excel.
5. Finish with an integrated marketing-and-risk case.

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Retrieval check

- Is `Have Tried` categorical or numerical?
- Is `Income` categorical or numerical?
- What does one row of the Marketing worksheet represent?
- Why should a business question come before an Excel calculation?

Probability measures uncertainty

- An **experiment** is a process with an uncertain outcome.
- The **sample space** S contains all possible outcomes.
- An **event** is one or more outcomes of interest.
- $0 \leq P(A) \leq 1$ and $P(S) = 1$.
- Probabilities can be expressed as proportions or percentages.

AND, OR, and NOT

Business wording	Notation	Meaning
A **and** B	$A \cap B$	both occur
A **or** B	$A \cup B$	at least one occurs
not A	A^c	complement of A

$$P(A^c) = 1 - P(A)$$

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

Mutually exclusive is not independent

- **Mutually exclusive:** A and B cannot occur together, so $P(A \cap B) = 0$.
- **Independent:** learning that B occurred does not change the probability of A.
- Non-trivial mutually exclusive events are dependent: observing one rules out the other.

Watch me: translate a business question

For an online order, define:

- A: delivery is late;
- B: customer requests a refund.

Translate:

1. “late and refunded” as $A \cap B$;
2. “late or refunded” as $A \cup B$;
3. “not late” as A^c; and
4. “refunded given late” as $P(B \mid A)$.

Now you try: events and rules

Time allowed: 4 minutes

Let C mean a customer clicks an advertisement and P mean they purchase.

1. Translate “clicked but did not purchase”.
2. Translate “clicked or purchased”.
3. If $P(C) = 0.30$, $P(P) = 0.12$, and $P(C \cap P) = 0.08$, calculate $P(C \cup P)$.
4. Explain why click and purchase are not mutually exclusive.

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Two categorical variables

A contingency table cross-classifies observations by two categorical variables.

- **Cell:** joint frequency or probability.
- **Margin:** total for one variable.
- **Grand total:** all observations.
- Percent of grand total supports joint and marginal probability.
- Row or column percentages support conditional probability.

Watch me: build the Marketing table

Open `week2-categorical-data.xlsx` and select **Marketing**.

Create a PivotTable:

1. put `Income` in Rows;
2. put `Have Tried` in Columns;
3. count `Person ID` in Values; and
4. show the counts as **% of Grand Total**.

Check that all cells sum to 100%.

Joint and marginal probability

- A **joint probability** refers to two events together: $P(A \cap B)$.
- A **marginal probability** refers to one variable and is read from a row or column total.

Always state the denominator:

$$P(A) = \frac{\text{observations satisfying } A}{\text{all observations}}.$$

Now you try: marketing probabilities

Time allowed: 7 minutes

Using the PivotTable, calculate and label:

1. $P(\text{not tried})$;
2. $P(\text{income} > 50{,}000 \cap \text{not tried})$;
3. $P(\text{income} > 50{,}000 \cup \text{not tried})$.

For each answer, identify whether it is marginal, joint, or union probability.

Conditional probability

Conditional probability changes the denominator:

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}.$$

Read $P(A \mid B)$ as “the probability of A, given B”. The event after the vertical bar defines the group being considered.

Watch me: condition on “not tried”

To calculate

$$P(\text{income} > 50{,}000 \mid \text{not tried}),$$

either:

- divide the relevant joint probability by $P(\text{not tried})$; or
- show PivotTable values as **% of Column Total** if `Have Tried` is in columns.

Verify that each conditioning column sums to 100%.

Now you try: reverse the condition

Time allowed: 6 minutes

Calculate both:

1. $P(\text{not tried} \mid \text{income} > 50{,}000)$;
2. $P(\text{income} > 50{,}000 \mid \text{not tried})$.

Explain why these are different questions. Which denominator is used in each calculation?

Independence

Events A and B are independent when any equivalent condition holds:

$$P(A \mid B) = P(A),$$

$$P(B \mid A) = P(B),$$

$$P(A \cap B) = P(A)P(B).$$

A small difference may arise from sampling variation; it is not

Watch me: tried product and gender

Create a second PivotTable for `Gender` and `Have Tried`.

Compare:

$$P(\text{tried} \mid \text{female}) \quad \text{with} \quad P(\text{tried}).$$

Narrate the conclusion as an association in this sample, not a causal effect.

Now you try: assess independence

Time allowed: 7 minutes

Use `Live Alone` and `Have Tried`.

1. Construct the contingency table.
2. Compare an appropriate conditional probability with its marginal probability.
3. Decide whether independence looks plausible.
4. Write one sentence that avoids causal language.

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From data to a probability model

- A histogram describes observed data.
- A probability density curve models possible values of a continuous variable.
- Probability is represented by **area under the curve**.
- Total area is 1.
- For a continuous variable, probability at one exact value is effectively 0.

The Normal distribution

If $X \sim N(\mu, \sigma)$, the distribution is:

- symmetric and bell-shaped;
- centred at mean μ; and
- spread according to standard deviation σ.

Approximately 68%, 95%, and 99.7% lie within one, two, and three standard deviations of the mean.

Excel's Normal functions

Question	Excel function
$P(X \leq x)$	`NORM.DIST(x, mean, sd`
$P(X > x)$	`1-NORM.DIST(x, mean,`
percentile with lower-tail probability p	`NORM.INV(p, mean, sd)`
standard Normal percentile	`NORM.S.INV(p)`

For an interval, subtract two cumulative probabilities.

Watch me: blood pressure risk

Assume adult systolic blood pressure is Normal with mean 124 and standard deviation 10.

- Below 117: `=NORM.DIST(117,124,10,TRUE)`
- Above 140: `=1-NORM.DIST(140,124,10,TRUE)`
- Highest 15% cutoff: `=NORM.INV(0.85,124,10)`

Sketch and shade the requested area before typing the formula.

Now you try: service time

Time allowed: 7 minutes

Assume service time is Normal with mean 8 minutes and standard deviation 1.5 minutes.

1. Find $P(X < 6)$.
2. Find $P(X > 10)$.
3. Find $P(7 < X < 9)$.
4. Find the 90th percentile.
5. For each, sketch the relevant area and write the Excel formula.

Standardising a value

The standard score

$$z = \frac{x - \mu}{\sigma}$$

counts how many standard deviations x is above or below the mean.

- $z > 0$: above the mean.
- $z < 0$: below the mean.
- Magnitude describes distance from the mean in standard-deviation units.

Now you try: compare unlike measures

Time allowed: 5 minutes

A store's weekly sales are 1.4 standard deviations above the chain mean. Its profit is 0.6 standard deviations above the chain mean.

1. Which result is more unusual relative to its distribution?
2. Why can standard scores compare variables measured in different units?
3. Does the comparison prove the store is well managed?

Student's t distribution

- The t distribution is symmetric and centred at zero.
- It has heavier tails than the standard Normal distribution.
- Its shape depends on degrees of freedom, commonly $df = n - 1$.
- It is used when estimating a mean and the population standard deviation is unknown.
- As df increases, t approaches the standard Normal distribution.

Watch me: choose the distribution

Ask in order:

1. Are we modelling individual values or a sample statistic?
2. Is a Normal model for the population stated or defensible?
3. Is the population standard deviation known?
4. If estimating a mean with unknown σ, what are the degrees of freedom?

The method follows the information available—not the desired answer.

Now you try: method selection

Time allowed: 5 minutes

Choose Normal, standard Normal, *t*, or “not enough information”:

1. Find a percentile of stated Normal delivery times.
2. Standardise one delivery time when μ and σ are known.
3. Estimate a population mean from $n = 18$ observations when σ is unknown.
4. Model a severely skewed individual outcome with no distributional assumption.

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Watch me: from question to conclusion

For “Are high-income respondents more likely to have tried the product?”:

1. define the two events;
2. build the correct PivotTable;
3. choose conditional probabilities with clear denominators;
4. compare them; and
5. state the sample association and its limitation.

The conclusion should answer the business question, not narrate menu clicks.

Now you try: marketing evidence brief

Time allowed: 10 minutes

Using the Marketing worksheet, choose one demographic variable and answer:

1. Is product trial associated with that variable?
2. Which joint, marginal, and conditional probabilities support your answer?
3. Is independence plausible?
4. What can and cannot be concluded from these observational data?

Prepare a three-sentence briefing and one clearly labelled

Common traps

- Treating “and” as addition without subtracting overlap.
- Reversing $P(A \mid B)$ and $P(B \mid A)$.
- Comparing conditional probabilities with different, hidden denominators.
- Calling mutually exclusive events independent.
- Using `NORM.DIST` without checking the tail or interval requested.
- Treating a model assumption as an empirical fact.

Closing check

Can you:

- translate business language into event notation?
- distinguish joint, marginal, union, and conditional probability?
- test the idea of independence using a contingency table?
- choose the correct Excel Normal function and tail?
- explain when a *t* distribution becomes relevant?
- communicate association without claiming causation?